

Text Extraction and Translation from an Image

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Abstract— Extracting text from images is a pivotal process in modern data management, facilitating the conversion of visual information into editable digital text. This paper investigates ways to speed up this conversion and provides a way to translate this text into the desired language. We focus on creating a text translator with audio capabilities, using OCR, Tesseract, and Google API for multilingual support. The study tackles the problems associated with storing hard copies of documents by using optical character recognition (OCR) systems. We show how printed characters from images can be recognized and converted into editable text, making data retrieval processes faster. Additionally, the OCR technology enhances the human-machine interface across various applications, providing scalability and accessibility, especially for users facing language barriers.

Index Terms— Text extraction, Optical Character Recognition (OCR), Tesseract, DPI (Dots per inch), Google API.

I. INTRODUCTION

[1] Character recognition is an intriguing combination of recognition of patterns and artificial intelligence. Optical Character Recognition (OCR) automatically adds pertinent information to an electronic database, eliminating the need for human retyping. Optical character recognition has several uses, including invoice imaging, legal, finance, and healthcare industries. Applications of OCR include Captcha, digital libraries, institutional repositories, number plate recognition, optical music recognition, and handwriting recognition [11]. It greatly enhances automation and improves human-machine interfaces in various applications. Several research efforts have focused on new strategies and procedures to improve recognition accuracy and minimize processing time. [3] Documents may be scanned as images, edited, and searched for further processing.

[11] OCR software aims to identify text and transform it into a modifiable format. Therefore, the main goal of OCR is to create computer algorithms that can detect the characters in the text. An optical scanner scans a document, creating a non-editable picture of it in the process. This text picture has been translated into editable character codes, like ASCII.

[7] This project aims to translate the text and apply text extraction from the image. In several image-based applications, including assistance navigation, supplementary reading, picture retrieval, scene comprehension, etc., scene images can act as indicative markings. [6] Compared to scanned documents, retrieving text from a real-life scene

images is a more difficult task due to the intricate backdrops and wide variations in text trends, including font, color, scale, intensity, and orientation. Then magnetic stripes have made some on-card payments safer but still leave card-not-present frauds at higher rates. Even then, there are chances for thieves to misuse credit cards. [10] There are many machine learning techniques to overcome this problem.

This work aims to show that text and interactive graphics may have a close dynamic relationship. [8] This kind of extraction may be shown by the camera on an Android handset, and the method can be simply implemented in Java. [8] Millions of mobile users worldwide carry their phones around all the time so they can easily take pictures and extract text.

A. Need and Motivation

The ability to convert images into editable text has become increasingly valuable in the modern digital era, where visual content is abundant. An easy and effective way to convert printed or handwritten text contained in images into editable and searchable text is by using our project. OCR technology is used by an image-to-text converter online, an online tool that extracts text from images. OCR algorithms translate characters in an image into machine-readable text by analyzing the contours, patterns, and shapes of the characters. This text can then be searched, edited, and shared, offering a flexible way for various online platforms.

B. Contributions

This paper aims to convey that image extraction and translation play a vital role in today's world. By converting visual content into editable and searchable text, an image-to-text converter online is an effective tool that helps visual content reach its full potential.

C. Organization of this paper

The remaining section of the paper is organized as follows. Section II provides a background and introduces key terms related to the field. Section III presents a literature review, examining existing research pertaining to the domain of text extraction and translation from an image. In Section IV, we present potential ideas for future work. Section V will explore the opportunities for applications in various domains. Finally, Section VI concludes the paper by summarizing our findings.

II. OVERALL BACKGROUND

Visitors to distant nations, whether for vacation or business, often need help to read and understand the local language. This practice improves students' ability to read and comprehend native language texts.

The main challenges involved in the text extraction process are:

1. One of the initial problems of text extraction is ensuring that the text data we are dealing with is high-quality and appropriate for the study. Text data may be noisy, inconsistent, incomplete, or contain spelling issues, grammatical faults, slang, acronyms, or several languages.
2. Finding terminology in one's language that is as close to the meaning of particular words as possible is one of the challenges that literary translators encounter.

We propose a system that can overcome the above challenges to the greatest extent.

A. Problem Statement

To create a system that recognizes text from natural photos and translates text into a language that the user has defined. The essential features of Extraction and Translation of text are- The most important algorithm, OCR, is used in our project. Character recognition for scripts involves text extraction, categorization, and post-processing. Typical text extraction includes a sequence of image transformations. The software uses segmentation, linked component analysis, and feature extraction techniques to reduce noise, distortion, and skewness from raw images and identify text segments that double-check the data, at least for training the model. Our approach is adaptable to various text formatting, such as font size, style, color, and backdrop. The suggested solution leverages complementary techniques, including effective character identification and localization, and a multiclass classifier to accurately recognize text.

III. LITERATURE SURVEY

3.1 Text Extraction and Detection from Images Using Machine Learning Techniques: A Research Review [2]

With prior knowledge and expertise, it can learn on its own without explicit programming or human help [2]. Mostly, it can predict how to complete some tasks or make judgments on its own using the training set and predicted input information. The practical uses of machine learning are numerous. An important application that retrieves significant information from images collected from different locations is text detection and extraction [2]. In images with complex backgrounds, text varies in terms of size, alignment, style, low brightness, and contrast. Because of the differences in formatting, many individuals find it difficult to read. This makes text extraction and detection increasingly important yet challenging jobs. Helping people from all over the world who speak different languages read and comprehend written languages is the aim of this project. Researchers have used diverse machine learning methodologies and instruments to recognize handwritten text and text extracted from images, then transform it into a digital format [1]. With the use of machine learning technology called optical character recognition (OCR), we can extract textual data or information from documents and convert it into searchable, editable data [2]. This study focuses on some machine learning techniques that can extract text from handwritten documents and photos, detect them in digital format, and translate them according to the user's requirements.

Advantage: -

This paper concludes that different technologies can recognize text using a variety of approaches. All of these previous works have provided valuable insights and discoveries.

Disadvantage: -

The disadvantage of this paper is that no full attention is given to translation. Old OCR systems have been used instead of new technologies.

3.2 Image Text Extraction and Its Language Translation.

Optical character recognition (OCR) translates written or printed text images into computer representations [1]. OCR comprises of the following steps: image acquisition, preprocessing, segmentation, feature extraction, and classification. Tesseract is an optical character recognition library for Python. The module reads text from images in JPEG, GIF, and PNG formats. Tesseract uses segmented to distinguish between the image's background and foreground and an adaptive recognition algorithm to match character pixels [3]. The primary purpose of OCR systems is to convert the contents of books, documents made of paper, newspapers, and other materials into an electronic format that is computer readable. Later, the data in digital form is suitable for many types of post-processing, such as translating languages and typeface modifications. Open CV is a Python package that allows us to do viewpoint transformations on images.

Several elements influence the quality of an image. The problems that arise mainly include noise, blur, uneven illumination, distortion, contrast, and resolution. In this study, we investigated and explored various image acquisition approaches for overcoming problems such as blur, low resolution, and uneven lighting conditions [3]. Additionally, they examined several methods and algorithms for character recognition in document photographs. It can translate document images into characters that may be recorded in a computer-readable electronic format and subsequently translated into any language that the Google Translate API supports.

Advantages: -

This paper uses the OpenCV library along with Google API for extraction and translation.

Disadvantages-

The main limitation is that only 'English' text can be extracted for translation.

3.3 Image to Text Converter and Translator using Deep Learning and Image Processing

This paper presents a specifically developed and trained neural network. The application extracts words independently for training purposes and maps them to the goal result. Automatic image processing is required to train an automatic classification engine for recognition purposes [3]. In modern times, smartphones are widely used to capture images. This technology aims to gather and translate words from input images into text. Text extraction from scanned documents can be challenging due to varied backgrounds and word patterns, encompassing scale, color, and size of fonts. Text recognition and extraction are essential for extracting texture from smartphone photos, as they identify regions with text characters [4]. After generating a picture using a smartphone camera, it undergoes multiple procedures comprising text detection, extraction, and translation. This paper describes a web program that employs Wikipedia to create a dictionary and extract terms [6]. Other programs like Google Translate, Mobile OCR, and Simple OCR have equivalent capabilities. However, text detection is not accurate. This method employs an implementation that incorporates all the details above, everything mentioned above. By employing keywords to extract all relevant text from Wikipedia, they have created dictionaries. A distinct value was given to every dictionary key. In all five languages, this value was compared to the text that was taken from the picture. The probability of a distributional graph was generated for every match.

Advantages: -

They have incorporated a language classification in addition to text identification. A model that calculates the proportion of words in a text that are native to one language was developed using machine learning.

Disadvantages: -

It takes a lot of time to translate the extracted text. Only five languages were taken for translation.

3.4 Extraction of Text from an Image and its Language Translation Using OCR.

Obtaining text from photos is a resilient process when using digital image processing. Detecting and recognizing text from images is a complicated task. Computer software can use complex algorithms to extract text from pictures. Therefore, it cannot be used in the current setting. Various sorts of language translators are available, including voice and keyboard-based. However, these interpreters are

difficult to use [5]. This work exhibits a dynamic relationship between text and animated images. The Android smartphone camera can perform this type of extraction, and the method can be readily implemented in Java. There are millions of mobile users. The technology can recognize characters with high accuracy, leading 90%. The advantages of this software's flexibility and flexibility make it suitable for detecting and translating multiple languages. Accurate recognition depends on the material's structure and quality. Advantages: -

An Android application was developed where a mobile camera captures the image/text in real-time.

Disadvantages: -

The point to be noticed here is that the implementation of the translation process was difficult. Only the recognition part was working out well.

3.5 Text Extraction and Detection from Images Using Machine Learning Techniques: A Research Review

A subfield of artificial intelligence, machine learning is seeing rapid global research expansion. It can learn independently, devoid of explicit programming or human assistance, thanks to its prior experience and knowledge. Using the relevant input dataset and its training set, it can make judgments on its own or make predictions while carrying out certain tasks [1]. Many real-time applications in our daily lives can benefit from machine learning. One of the crucial applications that uses significant information from photos taken from different sources is text extraction and detection. Text in diverse variations varies in low brightness or contrast in photos with complicated backgrounds, as well as in size, direction, alignment, and style several textual variations in text on images. As a result, text detection and extraction are more crucial and difficult tasks. The goal here is to assist people who speak different languages from all around the world in reading and understanding any language that is written. Researchers have used a variety of machine learning techniques and tools to identify handwritten text and text collected from photos and convert it to digital representation [1, 10]. Optical Character Recognition (OCR) is a machine learning approach that allows us to recognize and extract text data or information from documents, converting it into editable and searchable data. The focus of this study article is on machine learning techniques that may be used to extract text from handwritten documents and photos, identify them in digital format, and translate them to suit the needs of the user.

The back propagation algorithm was used for both training and testing [2]. The back propagation algorithm trains the system through iterative processing and reduces the mean squared error between prediction and actual class by modifying the weighted values in backward direction.

Advantages: -

This paper deduces that various technologies can be used to recognize text using a combination of machine learning algorithms and deep learning algorithms.

Disadvantages: -

The disadvantage in this paper includes that not full attention is given in case of translation. Old systems of OCR have been used instead of new technologies.

3.6 Text Recognition from an Image

To accomplish high-speed data processing, analog data must be converted into digital data. Storing hard copies of paperwork takes up a lot of room and can be time-consuming to retrieve information. The optical character recognition system is a successful method to recognize printed characters. It allows you to easily recognize and convert written text on an image into editable text. It also speeds up data retrieval from images [4]. Images with characters can be scanned and converted into machine-readable characters using an OCR system. Character recognition is one of the most interesting areas of pattern recognition and artificial intelligence. Optical Character Recognition extracts the relevant information and automatically enters it into electronic database instead of the conventional way of manually retyping the text. Optical Character Recognition is a vast field with several varied applications such as invoice imaging, legal industry, banking, health care industry etc. [1]. OCR is also widely utilized in many other fields like Captcha, Institutional repositories and digital libraries, Optical Music Recognition without any human correction or human effort, Automatic number plate identification and Handwritten Recognition [6]. It makes significant contributions to the advancement of an automation process and has the potential to improve the interface between man and machine in a variety of applications. Several research studies have focused on novel strategies and methodologies that would cut processing time while improving recognition accuracy. It is now possible to scan documents as images and make them modifiable and visible for further information processing. The goal of OCR software is to identify the text and then convert it to customizable format. Thus, the primary goal of OCR is to create computer algorithms that can detect characters in the text [11]. An optical scanner first scans a document, creating an image that cannot be edited. Optical character recognition includes translation of this text image into editable character codes, such as ASCII.

Advantages: -

Important feature mainly OCR was used.

Disadvantages: -

Restriction of languages is observed in this paper.

IV PROPOSED METHODOLOGY

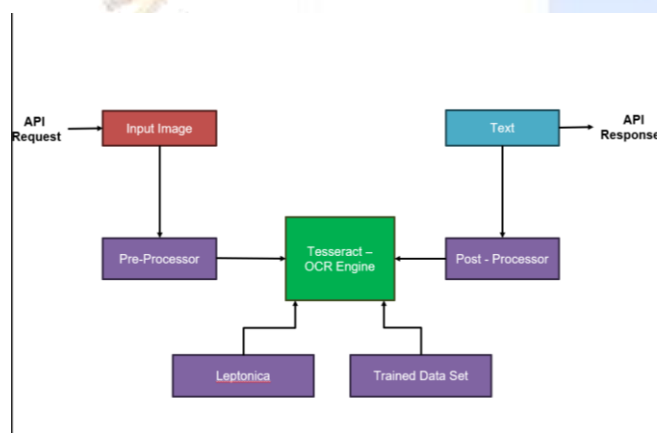


Fig (a)

OCR Tesseract workflow

The Tesseract OCR Workflow depicts the overall process API Request and Response. Pre- Processor, Post-Processor, Trained Data set, Leptonica are involved in the whole architecture.

Leptonica is an academically focused open-source library

That includes tools for image-processing and for analysis Applications.

- Our suggested system extracts text from an image using machine learning techniques on a grid infrastructure, resulting in a character recognition technique that can recognize characters from multiple languages.
- This feature, referred to as grid infrastructure, overcomes the problem of diverse character recognition and supports different functionalities to be performed on the image. In this context, Gridinfrastructure means the infrastructure that supports group of specific set of languages.
- Thus, extraction of text from an image using fuzzy logic on a grid infrastructure is multilingual.
- The suggested system addresses the shortcoming of the existing system by supporting different functionalities such as editing and searching.
- Improves recognition of diverse characteristics.

Steps: -

- (1) The process of scanning (reading) operation provides input to the system.
- (2) The OCR technology is used to that read image. The tesseract module text is extracts text from the supplied image.
- (3) The retrieved text is able to be translated into the user's choice languages.

The Web-App will allow the user to attach an image and he/she will be able to choose any language they want to translate, then the translated text will be displayed.

Audio of the translated text will also be as an output in our project.

Tesseract works best on images which have a DPI (dots perinch) of at least 300 dpi, so it may be beneficial to resize images.

Files supported are jpeg, png, gif, bmp, tiff, and others. Rescaling, binarization, noise removal, deskewing, etc. areinvolved in the pre- processing of the OCR.

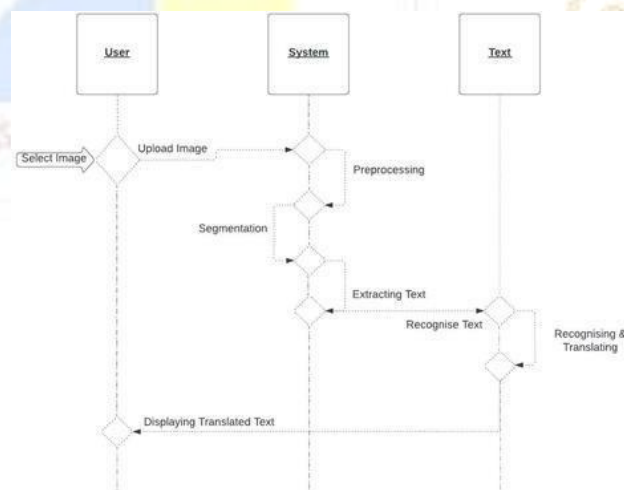


Fig (b)

Sequence Diagram

The above Sequence Diagram shows the co-relation between User, System, and Text. The 'System' is the OCR system here.

4.1 ALGORITHM

Optical Character Recognition (OCR) is a technology that turns printed, handwritten, and typed characters into machine-encoded text. OCR systems typically use a combination of image processing, pattern recognition, and machine learning approaches to identify and recognize characters in images. There are some key features of OCR:

Character Recognition: OCR's principal role is to identify and extract characters (letters, numerals, and symbols) from photos or scanned documents. It recognizes individual characters and turns them into editable, searchable text. **Preprocessing:** OCR frequently includes preprocessing steps to improve image quality prior to character recognition. This could include image resizing, binarization (converting grayscale to binary images), noise reduction, and skew correction.

Optical Character Recognition (OCR) transforms scanned documents into computer-readable text. The OCR procedure should result in an invoice structure that is similar to the original.

Steps:

1. Load any image format (bmp, jpg, png) from a specified source. To binarize an image, convert it to grayscale and apply the Otsu algorithm's threshold value.
2. //completed (How can I eliminate noise from the output image???)
3. Detecting image attributes such as resolution and inversion. Finally, we can straighten the image for further processing.
4. Line identification and removal. This phase improves pagelayout analysis, recognizes highlighted text, and detectstables.
5. Page layout analysis. In this stage, I aim to identify the textzones in the image. Only a fraction of the region is used for recognition, leaving out the rest.
6. Detecting text lines and words. It's important to consider font size and word spacing.
7. Recognition of characters. The main algorithm of OCR is converting an image of each character to its corresponding character code. This technique generates multiple character codes for ambiguous photos.

FUTURE SCOPE

1. Legal Industry

Legal practitioners can easily search through large datasets using keywords.

2. Healthcare.

Healthcare practitioners must manage several patient documents, including insurance and general health forms. To manage this information, it's helpful to store it in an electronic database for easy access. OCR- powered form processing solutions collect information from forms and store it in databases, ensuring quick recording of patient data information.

PSEUDO CODE

1. Input: Image containing text
2. Output: Translated text from the image
3. // Step 1: Text Recognition (OCR)
4. `captured_text = recognize_text_from_image(image)`

5. // Step 2: Language Translation

6. `translated_text = translate_text(captured_text, source_language, target_language)`

7. // Display or use translated_text as needed

8. `print(translated_text)`

// Function to perform OCR (Text Recognition) on the input image

`function recognize_text_from_image(image):`

// Use OCR library or service to extract text from the image

// Example: Tesseract OCR, Google Cloud Vision API, etc.

`text = OCR(image)`

`return text`

// Function to translate text from source language to target language

`function translate_text(text, source_language, target_language):`

// Use translation service or library to translate text

// Example: Google Translate API, Microsoft Translator API, etc.

`translated_text = translate(text, source_language, target_language)`

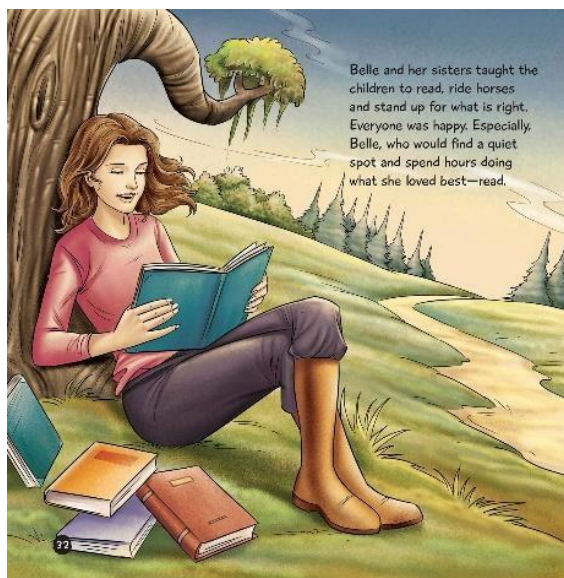
`return translated_text`

Step 1 (Text Recognition): This part focuses on using an OCR library or service to recognize text from the input image. We have used a specialized OCR tool Tesseract OCR, or a cloud-based service such as Google Cloud Vision API also can be used.

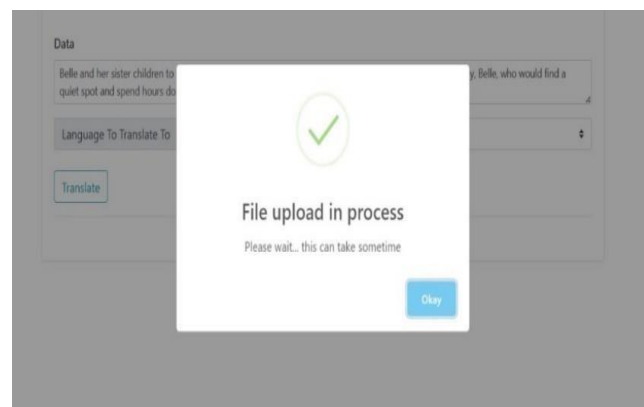
Step 2 (Language Translation): After extracting text from the image, the next step involves translating this text into the desired target language. We have used a translation

RESULTS

1. ENGLISH TEXT



Fig(c)



Fig(f)

Data has been shown, SELECT a particular language in which the text gets translated. After selecting the upload of file will start.

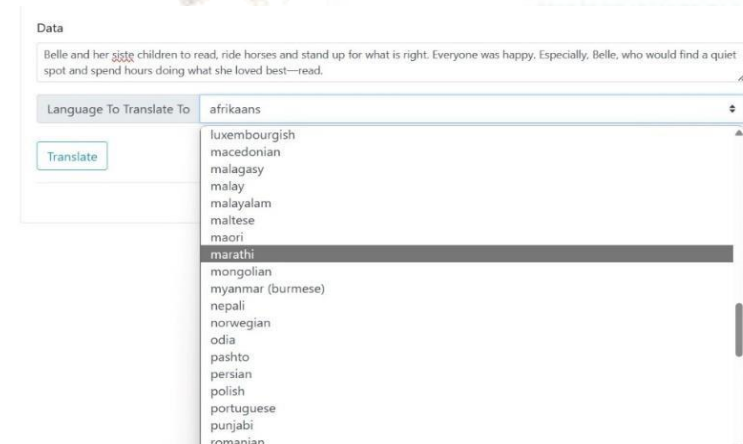


Fig(g)

In the above screenshot, the English text has been translated in 'Marathi' language selected by the user.

2. SPANISH IMAGE

In the above window the user can upload an image with text that needs to be translated. In this, we have attached an English image text.



Fig(e)



Fig(h)



Fig(i)



Fig(l)



Fig(j)

The text in the image Fig (h) has been translated in 'Turkish' And 'Afrikaans' language with audio feature of the respective Translated text.

3. BLUR IMAGE

In this project blur image's text can also be translated in any user-defined language.

In Fig(k) the text can be extracted and translated even if it is a blur image.



Fig(k)

The text in Fig (k) is translated in 'German' language with the Following Audio.

We, propose a system in the user can extract text from any image and text can be translated into any user-defined language With audio feature.

VI CONCLUSION

This topic focuses on optical character recognition approaches for translating text from unknown languages into recognized languages. The system has the ability to recognize characters with an accuracy rate greater than 90%. This system's advantages include portability, scalability, and the capacity to recognize and translate multiple languages. Accurate recognition depends on the material being read and its level of accuracy. Our system's performance is measured by the number of successful recognitions. We chose recognition rate over successful translation rate performance criterion because recognition rate directly measures the success of the text identification algorithm, whereas translation rate is influenced by the Google translation engine, over which we have no control. The user will receive the translated text from the image provided.

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